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Display panel and display terminal

Abstract

The present application provides a display panel and a display terminal. The display panel includes a first metal layer and a second metal layer. The first metal layer includes data lines arranged in a display area and extending along a first direction. The second metal layer includes first transmission lines in the display area. Each of the first transmission lines is electrically connected to at least one data line. At least a portion of each of the first transmission lines extends along the first direction.

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Background/Summary

RELATED APPLICATIONS

(1) This application is a National Phase of PCT Patent Application No. PCT/CN2022/101445 having International filing date of Jun. 27, 2022, which claims the benefit of priority of Chinese Patent Application No. 202210604054.3 filed on May 30, 2022. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

FIELD AND BACKGROUND OF THE DISCLOSURE

(2) The present application relates to a field of display technology and in particular, to a display panel and a display terminal.

(3) With the development of display panel technology, screen ratios have become an important criterion for product differentiation. In development of display panels today, there is an increasing trend towards a full screen. In the display panel design, narrowing a bezel width can greatly increase a screen ratio, so a narrow bezel design has attracted much attention. Traditional display panels generally have a first gate layer and a second gate layer in horizontal wiring to transmit gate driving signals, and have a first source-drain layer and a second source-drain layer in vertical wiring to transmit data signals and power signals. Furthermore, the traditional display panels have fan-out lines in left and right arc regions at a lower bezel of the display panel to connect data lines in a display area to a non-display area. The fan-out lines generally include the first gate layer and the second gate layer arranged alternately. The non-display area is a channel for all driving signal traces. A spacing of the data signal traces in the non-display area is much smaller than a spacing of the data lines in the display area, and there are other signal traces in the non-display area, so a layout design of the other signal traces and the data signal traces in the non-display area leads to a height of the fan-out traces, which occupies a large space at the lower bezel, resulting in a wider lower bezel.

(4) In order to satisfy the need for narrower bezels, a space for wiring is getting smaller, and a spacing between lines is almost close to a smallest possible spacing in a production line. Even if the design is based on an extremely small spacing, requirements for narrow bezel requirements cannot be satisfied. Therefore, it is necessary to improve this defect.

SUMMARY OF THE INVENTION

(5) The present application provides a display panel, which is used to solve a technical problem that lower bezels of conventional display panels are too wide and cannot meet design requirements for narrow bezels.

(6) The present application provides a display panel. The display panel includes a display area and a non-display area arranged on at least one side of the display area, wherein the display panel includes a substrate, a first metal layer, and a second metal layer. The first metal layer is disposed on one side of the substrate, wherein the first metal layer includes a plurality of data lines disposed in the display area and extending along a first direction. The second metal layer is disposed on one side of the first metal layer away from the substrate, and the second metal layer includes a plurality of first transmission lines in the display area. Each of the first transmission lines is electrically connected to one of the data lines, and at least a portion of each of the first transmission lines extends along the first direction. A distance between each two adjacent first transmission lines extending along the first direction is smaller than a distance between each two adjacent data lines.

(7) The present application further provides a display terminal. The display terminal includes a terminal body and a display panel. The terminal body and the display panel are combined into one, and the display panel includes a display area and a non-display area located on at least one side of the display area. The display panel includes: a substrate, a first metal layer, and a second metal layer. The first metal layer is disposed on one side of the substrate, wherein the first metal layer includes a plurality of data lines disposed in the display area and extending along a first direction. The second metal layer is disposed on one side of the first metal layer away from the substrate, wherein the second metal layer includes a plurality of first transmission lines in the display area,

each of the first transmission lines is electrically connected to one of the data lines, and at least a portion of each of the first transmission lines extends along the first direction. A distance between each two adjacent first transmission lines extending along the first direction is smaller than a distance between each two adjacent data lines.

Advantages of the Present Application

(8) A display panel of the present application includes a display area and a non-display area located on at least one side of the display area, the display panel includes a substrate, a first metal layer, and a second metal layer. The first metal layer is disposed on one side of the substrate. The first metal layer includes a plurality of data lines disposed in the display area and extending along a first direction. The second metal layer is disposed on one side of the first metal layer away from the substrate. The second metal layer includes a plurality of first transmission lines in the display area, and each of the first transmission lines is electrically connected to one of the data lines. At least a portion of each of the first transmission lines extends along the first direction. A distance between each two adjacent first transmission lines extending along the first direction is smaller than a distance between each two adjacent data lines. In the present application, by arranging the first transmission lines in the second metal layer and by transmitting data signals from the non-display area to the data line in the display area through the first transmission lines, fan-out lines can be arranged in the display area (Fan-out in AA). A distance between each two adjacent first transmission lines extending along the first direction is smaller than a distance between each two adjacent data lines, so that the first transmission lines can be arranged more densely, thus reducing an overall width occupied by the first transmission lines extending along the first direction, reducing and narrowing left and right arc regions at the lower bezel. That is to say, the present application can realize an extremely narrow frame design without adding extra processes or film layers.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) FIG. 1 is a schematic view of a basic structure of a display panel according to one embodiment of the present application.
- (2) FIG. 2 is a schematic view of a film layer structure of the display panel according to one embodiment of the present application.
- (3) FIG. 3 is a schematic view of a wiring arrangement of first transmission lines according to one embodiment of the present application.
- (4) FIG. 4 is a schematic view of an electrical connection between a first transmission section and a data line according to one embodiment of the present application.
- (5) FIG. 5 is a schematic view of a film layer structure of an active layer according to one embodiment of the present application.
- (6) FIG. 6 is a schematic view of a film layer structure of a first gate layer according to one embodiment of the present application.
- (7) FIG. 7 is a schematic view of a film layer structure of a second gate layer according to one embodiment of the present application.
- (8) FIG. 8 is a schematic view of a film layer structure of a first metal layer according to one embodiment of the present application.
- (9) FIG. 9 is a schematic view of a film layer structure of a second metal layer according to one embodiment of the present application.
- (10) FIG. 10 is a schematic view of a film layer structure of a sub-pixel according to one embodiment of the present application.
- (11) FIG. 11 is a schematic view illustrating a basic structure of the display panel according to

another embodiment of the present application.

DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

(12) Technical solutions of the present application will be clearly and completely described below with reference to the accompanying drawings and in conjunction with specific embodiments of the present application. In the drawings, for clarity and ease of understanding and description, dimensions and thicknesses of components depicted in the drawings are not to scale.

(13) FIG. 1, FIG. 2, FIG. 3, and FIG. 4 are a schematic view of a basic structure of a display panel, a schematic view of a film layer structure of the display panel, a schematic view of a wiring arrangement of first transmission lines, and a schematic view of an electrical connection between a first transmission section and a data line according to one embodiment of the present application, respectively. The display panel includes a display area A1 and a non-display area A2 located on at least one side of the display area A1. The display panel includes a substrate 11, a first metal layer 20, and a second metal layer 30. The first metal layer 20 is disposed on one side of the substrate 11. The first metal layer 20 includes a plurality of data lines Data disposed in the display area A1 and the extending along the first direction. The second metal layer 30 is disposed on one side of the first metal layer 20 away from the substrate 11. The second metal layer 30 includes a plurality of first transmission lines 301 in the display area A1. Each of the first transmission lines 301 is electrically connected to at least one of the data lines Data, and at least a portion of each first transmission line 301 extends along a first direction. A distance between two adjacent first transmission lines 301 extending along the first direction is smaller than a distance between two adjacent data lines Data.

(14) It can be understood that in conventional techniques, fan-out lines are usually arranged in the non-display area A2, so as to transmit data signals from the non-display area A2 to the data lines Data in the display area A1. However, a space occupied by the fan-out lines is relatively large, so that bezels of the display panel cannot be further reduced, and a design requirement for narrow bezels cannot be met. In the present application, the second metal layer 30 is used for a wiring arrangement of the first transmission lines 301. Through the first transmission lines 301, data signals are transmitted from the non-display area A2 to the data line Data in the display area A1, so as to realize a design in which the fan-out lines are placed in the display area A1. Moreover, the distance between two adjacent first transmission lines 301 extending along the first direction is smaller than the distance between two adjacent data lines Data. Consequently, the first transmission lines 301 can be arranged more densely, and the overall width occupied by all the first transmission lines 301 can be decreased. This further reduces and narrows left and right arc regions at the lower bezel of the display panel. That is to say, an extremely narrow bezel design can be realized without additional film layers and processes.

(15) In one embodiment, in the display area A1, along a second direction, in the same unit distance, a number of the first transmission lines 301 is greater than a number of the data lines Data, and the second direction is perpendicular to the first direction.

(16) It can be understood that, in the present embodiment, in the unit distance in the second direction, by making the number of the first transmission lines 301 greater than the number of the data lines Data, the first transmission lines 301 can be arranged more densely, and an overall width of the first transmission lines 301 can be reduced. Furthermore, the left and right arc regions in the bezel frame of the display panel can be reduced and narrowed. That is to say, an extremely narrow bezel design can be realized without additional film layers and processes.

(17) It should be noted that the first metal layer 20 is a film layer for transmitting source and drain signals, and the second metal layer 30 is located between the first metal layer 20 and an anode layer (not illustrated). Driving electrical signals generated by thin film transistors (not illustrated) are transmitted from the first metal layer 20 to the second metal layer 30 and then to the anode layer.

(18) It should be noted that conventional techniques use the first gate layer 60 or the second gate layer 70 for wiring of the fan-out lines. Instead, the present application uses the second metal layer

30 for wiring of the fan-out lines. Compared with the first gate layer **60** or the second gate layer **70**, the second metal layer **30** has lower resistance, and has a low degree of signal attenuation. In the present application, the second metal layer **30** is effectively used to arrange the first transmission lines **301**. Compared with conventional double-layer source-drain products, the present application does not need additional other film layers, nor does it need additional photomask processes. Thus, the present application prevents an increase in manufacturing costs, and a manufacturing process is simple, so that mass production can be realized.

(19) It should be noted that only the second metal layer **30** is shown in FIG. 1. In one embodiment, the first transmission line **301** includes a first transmission section **3011** and a second transmission section **3012** in the display area **A1**, the second transmission section **3012** extends along the first direction, one end of the first transmission section **3011** is electrically connected to the data line **Data**, and the other end of the first transmission section **3011** is electrically connected to the second transmission section **3012**.

(20) Specifically, the first transmission line **301** is divided into a first transmission section **3011** and a second transmission section **3012**. The second transmission section **3012** extends along the first direction (i.e., a vertical direction), and the first transmission section **3011** extends along the second direction (i.e., a horizontal direction). The second transmission section **3012** vertically transmits data signals upward from the non-display area **A2**. The first transmission section **3011** is connected to the second transmission section **3012**, and the first transmission section **3011** transmits horizontally the vertically transmitted data signal. The data signal of each first transmission line **301** is correspondingly transmitted to at least one data line **Data** in the display area **A1**.

(21) It should be noted that FIG. 2 only shows a wiring arrangement of one first transmission line **301**. It can be seen from FIG. 2 that the second transmission section **3012** extends in the first direction and is connected to the first transmission section **3011**. The first transmission section **3011** extends along the second direction and is electrically connected to the data line **Data**. That is to say, the data signal is transmitted through the second transmission section **3012** to the first transmission section **3011** (numeral reference 4.fwdarw.5), and is transmitted from the first transmission section **3011** to the data line **Data**. Then, the data signal can be transmitted through the data line **Data** to any sub-pixel **10** in an entire column. Accordingly, the present application realizes that the data signal is transmitted from the non-display area **A2** to the display area **A1**.

(22) It should be noted that reference numerals **1**, **2**, **3**, and **4** in FIG. 2 indicate the second transmission sections **3012** which are used to transmit the data signal vertically upward from the non-display area **A2**. The reference numeral **5** indicates the first transmission section **3011** which horizontally transmits the data signal of the second transmission section **3012** to the data line **Data**. Specifically, as shown in FIG. 3, the lines marked **1**, **2**, **3**, and **4** transmit data signals horizontally through the line marked **5**. Then, the data signals are transmitted to the data line **Data** through a through hole **40**, thereby realizing data signal transmission.

(23) It should be noted that the display panel further includes scan lines **Scan**. The scan lines **Scan** are arranged horizontally and are used to transmit gate driving signals. The gate driving signals are supplied from a lateral bezel.

(24) In one embodiment, a distance between two adjacent second transmission sections **3012** is smaller than the distance between two adjacent data lines **Data**. In this embodiment, it can be understood that, by making the distance between two adjacent second transmission sections **3012** smaller, the second transmission sections **3012** can be arranged more densely, and an overall width occupied by the second transmission sections **3012** can be decreased. Then, left and right arc regions at the lower bezel of the display panel can be narrowed and reduced. That is to say, an extremely narrow bezel design can be realized without additional processes and film layers.

(25) In one embodiment, in the display area **A1**, along the second direction, in the same unit distance, a number of the second transmission sections **3012** is greater than the number of the data lines **Data**. The second direction is perpendicular to the first direction.

(26) It can be understood that, in the present embodiment, by making the number of the second transmission sections **3012** per unit distance in the second direction greater than the number of the data lines Data, the second transmission sections **3012** can be arranged more densely, thus reducing the number of the second transmission sections **3012**. Accordingly, the overall width occupied by the second transmission sections **3012** can further narrow the left and right arc regions at the lower bezel of the display panel. That is to say, an extremely narrow bezel design can be realized without additional processes or film layers.

(27) In one embodiment, the display panel further includes a first insulating layer **21**. The first insulating layer **21** is disposed between the first metal layer **20** and the second metal layer **30**. The first insulating layer **21** includes a plurality of through holes **40**. One end of the first transmission section **3011** is electrically connected to the data line Data through the through hole **40**. That is to say, the data signal of the first transmission section **3011** is transmitted to the data line Data of the first metal layer **20** through the through hole **40** in the first insulating layer **21**.

(28) In one embodiment, the display area A1 is provided with multiple sub-pixels **10**. The sub-pixels **10** include a plurality of sub-pixel columns **100** arranged at intervals along the second direction. At least one of the sub-pixel columns **100** overlaps with at least two of the second transmission sections **3012**, and the second direction is perpendicular to the first direction.

(29) Specifically, as can be seen from FIG. 3, one sub-pixel column **100** overlaps with two second transmission sections **3012**. Therefore, the second transmission sections **3012** can be arranged more densely, thus reducing the overall width of the second transmission sections **3012**. This reduces and narrows the arc region (i.e., a lower left corner area in FIG. 3) at the lower bezel of the display panel, thereby realizing an extremely narrow bezel design.

(30) Continuing to refer to FIG. 1, in one embodiment, the second metal layer **30** includes at least one second transmission line **302** in the display area A1, the second transmission line **302** is electrically connected to the data line Data, and the second transmission line **302** extends along the first direction. The first transmission lines **301** are located at two sides of the second transmission line **302**.

(31) It can be understood that, in the present embodiment, the data signal can be transmitted to the data line Data through the first transmission line **301**, and can also be transmitted to the data line Data through the second transmission line **302**. The second transmission line **302** extends along the first direction. That is to say, the data signal is directly transmitted vertically to the through hole **40** through the second transmission line **302**, and does not need to be transmitted horizontally, which can reduce the difficulty in wiring and improve wiring flexibility.

(32) In one embodiment, the end of the first transmission section **3011** electrically connected to the data line Data is at one side of the second transmission line **302** and away from the end of the first transmission section **3011** electrically connected to the second transmission section **3012**. It can be understood that, by setting the first transmission section **3011** to transmit data signals in a direction away from the second transmission line **302**, the present embodiment prevents a short circuit between the first transmission section **3011** and the second transmission line **302**, which results in poor data signal transmission.

(33) In one embodiment, the first transmission lines **301** are symmetrically arranged with respect to the second transmission line **302**. It can be understood that, by arranging the first transmission lines **301** symmetrically with respect to the second transmission line **302** in this embodiment, the first transmission lines **301** on both sides of the second transmission line **302** can be evenly distributed, a length of the second transmission section **3012** in the first direction is reduced, and the impedance of the first transmission line **301** is reduced, thereby reducing the attenuation of the data signal.

(34) In one embodiment, the display panel includes a virtual central axis L0 along the first direction, and the second transmission line **302** coincides with the virtual central axis L0. It can be understood that in the present embodiment, by arranging the second transmission line **302** at the position of the virtual central axis L0, the space on the left and right sides of the second

transmission line **302** can be made equal. Accordingly, the present application can reserve the space close to the left and right bezels of the display panel, so as to narrow the space of the left and right arc regions at the lower bezel of the display panel, and meet the requirements for the narrow bezel design.

(35) In one embodiment, in the first direction, a length of each of the second transmission sections **3012** gradually decreases along a direction away from the second transmission line **302**. It should be noted that, in the first direction, the length of each of the second transmission sections **3012** gradually decreases along the direction away from the second transmission line **302**, so the present application can avoid a short circuit between the first transmission lines **301**. A length of each of the first transmission section **3011** gradually increases along the direction away from the second transmission line **302** to balance a difference resulting from the gradually shorter second transmission sections **3012**, so that impedances of the first transmission lines **301** are uniformly distributed, and a better transmission effect is achieved.

(36) In one embodiment, an orthographic projection of each of at least one of the second transmission sections **3012** projected on the first metal layer **20** coincides with one of the data lines Data. It can be understood that, by arranging the second transmission section **3012** above the data line Data, the present embodiment can shield an influence of jumping changes of the data signal on a gate (i.e., a Q point) of a driving transistor in a pixel circuit of the sub-pixel **10**, and can realize a wiring arrangement in which the second transmission line **302** is directly electrically connected to the data line Data through the through hole **40**.

(37) In one embodiment, the first metal layer **20** further includes a plurality of power supply lines VDD, the power supply lines VDD extend along the first direction. An orthographic projection of each of at least one of the second transmission sections **3012** projected on the first metal layer **20** coincides with one of the power supply lines VDD.

(38) It should be noted that the power supply lines VDD in conventional techniques are arranged in the second metal layer **30**. In the present embodiment, the power supply lines VDD and the data lines Data are arranged in the same layer to free up the second metal layer **30** and facilitate the wiring arrangement of the first transmission lines **301** and/or the second transmission line **302**. In addition, the power supply lines VDD are located inside the data lines Data. Therefore, by arranging the second transmission section **3012** above the power supply line VDD, the present application can also shield the influence of the jumping changes of the data signal on the gate (i.e. the Q point) of the driving transistor in the pixel circuit of the sub-pixel **10**.

(39) Continuing to refer to FIG. 2, it should be noted that the sub-pixels **10** provided in the present application have a 7T1C pixel structure. A 7T1C pixel circuit connection configuration is not the main point of the present application, and therefore a detailed description of the 7T1C pixel circuit connection configuration is omitted. Specifically, the display panel further includes an active layer **50**, a first gate layer **60**, and a second gate layer **70**. Please refer to FIG. 5 to FIG. 10, which are schematic film layer structure views showing the active layer, the first gate layer, the second gate layer, the first metal layer, the second metal layer, and the sub-pixel. The display panel further includes a light-emitting unit **80**.

(40) In one embodiment, the display panel includes the first gate layer **60** and the second gate layer **70**. The first gate layer **60** is located at one side of the first metal layer **20** and away from the second metal layer **30**. The second gate layer **70** is located between the first gate layer **60** and the first metal layer **20**. One end of the first transmission line **301** close to the non-display area A2 is electrically connected to the first gate layer **60** or the second gate layer **70**.

(41) Specifically, please refer to FIG. 11, which is a schematic view illustrating a basic structure of the display panel according to another embodiment of the application. It should be noted that, in this embodiment, the second transmission section **3012** of the first transmission line **301** only transmits data signals vertically upward from a bottom of the display area A1, and the data signals are transmitted from the non-display area A2 to the bottom of the display area A1 through fan-out

lines 90. The fan-out lines 90 are alternately arranged through the first gate layer 60 and the second gate layer 70. As can be seen from FIG. 11, by using the wiring arrangement of this embodiment, the space of left and right arc regions A3 at the lower bezel of the display panel can be reduced and narrowed, thereby realizing the extremely narrow bezel design.

(42) The present application further provides a display terminal, including a terminal body and the above-mentioned display panel. The terminal body and the display panel are combined into one. Please refer to FIG. 1 to FIG. 11 and related descriptions for the structure of the display panel, and a detailed description thereof is not repeated here. The display terminal of the present application can be a product or component with a display function, such as a mobile phone, a tablet computer, a notebook computer, a television, a digital camera, and a navigator.

(43) The display panel and the display terminal of the present application are described in detail above. It should be understood that the exemplary embodiments described herein are illustrative only, and are used for ease of understanding the method and the main ideas of the present application, but not to limit the present application.

Claims

1. A display panel comprising a display area and a non-display area arranged on at least one side of the display area, wherein the display panel comprises: a substrate; a first metal layer disposed on one side of the substrate, wherein the first metal layer comprises a plurality of data lines disposed in the display area and extending along a first direction; a second metal layer disposed on one side of the first metal layer away from the substrate, wherein the second metal layer comprises a plurality of first transmission lines in the display area, each of the first transmission lines is electrically connected to one of the data lines, and at least a portion of each of the first transmission lines extends along the first direction a first gate layer disposed on one side of the first metal layer away from the second metal layer; and a second gate layer disposed between the first gate layer and the first metal layer; wherein a distance between each two adjacent first transmission lines extending along the first direction is smaller than a distance between each two adjacent data lines, wherein one end of each of the first transmission lines close to the non-display area is electrically connected to the first gate layer or the second gate layer.
2. The display panel according to claim 1, wherein each of the first transmission lines comprises a first transmission section and a second transmission section in the display area, each of the second transmission sections extends along the first direction, one end of each of the first transmission sections is electrically connected to one of the data lines, and another end of each of the first transmission sections is electrically connected to one of the second transmission sections.
3. The display panel according to claim 2, wherein the second metal layer comprises at least one second transmission line disposed in the display area, the at least one second transmission line is electrically connected to at least one of the data lines, and the at least one second transmission line extends along the first direction; wherein the first transmission lines are disposed at two sides of the at least one second transmission line.
4. The display panel according to claim 3, wherein in each first transmission section, the end of the first transmission section electrically connected to one of the data lines is at one side of the at least one second transmission line and away from the end of the first transmission section electrically connected to one of the second transmission sections.
5. The display panel according to claim 4, wherein the first transmission lines are symmetrically arranged with respect to the at least one second transmission line.
6. The display panel according to claim 3, wherein the display panel comprises a virtual central axis along the first direction, and one of the at least one second transmission line coincides with the virtual central axis.
7. The display panel according to claim 3, wherein in the first direction, a length of each of the

second transmission sections gradually decreases in a direction away from the at least one second transmission line.

8. The display panel according to claim 2, wherein a distance between each two adjacent second transmission sections is smaller than a distance between each two adjacent data lines.

9. The display panel according to claim 2, wherein in the display area, along a second direction, per same unit distance, a number of the second transmission sections is greater than a number of the data lines, and the second direction is perpendicular to the first direction.

10. The display panel according to claim 2, wherein the display panel further comprises: a first insulating layer arranged between the first metal layer and the second metal layer, wherein the first insulating layer comprises a plurality of through holes, and one end of each of the first transmissions is electrically connected to one of the data lines through one of the through holes.

11. The display panel according to claim 2, wherein a plurality of sub-pixels are disposed in the display area, the sub-pixels comprise a plurality of sub-pixel columns spaced along the second direction, at least one of the sub-pixel columns overlaps at least two of the second transmission sections, and the second direction is perpendicular to the first direction.

12. The display panel according to claim 2, wherein an orthographic projection of each of at least one of the second transmission sections projected on the first metal layer coincides with one of the data lines.

13. The display panel according to claim 2, wherein the first metal layer further includes a plurality of power supply lines, the power supply lines extend along the first direction, and an orthographic projection of each of at least one of the second transmission sections projected on the first metal layer coincides with one of the power supply lines.

14. The display panel according to claim 1, wherein in the display area, along a second direction, per same unit distance, a number of the first transmission lines is greater than a number of the data lines, and the second direction is perpendicular to the first direction.

15. A display terminal, comprising a terminal body and a display panel, wherein the terminal body and the display panel are combined into one, and the display panel comprises a display area and a non-display area located on at least one side of the display area; wherein the display panel comprises: a substrate; a first metal layer disposed on one side of the substrate, wherein the first metal layer comprises a plurality of data lines disposed in the display area and extending along a first direction; a second metal layer disposed on one side of the first metal layer away from the substrate, wherein the second metal layer comprises a plurality of first transmission lines in the display area, each of the first transmission lines is electrically connected to one of the data lines, and at least a portion of each of the first transmission lines extends along the first direction; a first gate layer disposed on one side of the first metal layer away from the second metal layer; and a second gate layer disposed between the first gate layer and the first metal layer; wherein a distance between each two adjacent first transmission lines extending along the first direction is smaller than a distance between each two adjacent data lines, wherein one end of each of the first transmission lines close to the non-display area is electrically connected to the first gate layer or the second gate layer.

16. The display terminal according to claim 15, wherein each of the first transmission lines comprises a first transmission section and a second transmission section in the display area, each of the second transmission sections extends along the first direction, one end of each of the first transmission sections is electrically connected to one of the data lines, and another end of each of the first transmission sections is electrically connected to one of the second transmission sections.

17. The display terminal according to claim 16, wherein the second metal layer comprises at least one second transmission line disposed in the display area, the at least one second transmission line is electrically connected to the data lines, and the at least one second transmission line extends along the first direction; wherein the first transmission lines are disposed at two sides of the at least one second transmission line.

18. The display terminal according to claim 17, wherein the display panel comprises a virtual central axis along the first direction, and one of the at least one second transmission line coincides with the virtual central axis.

19. The display panel according to claim 17, wherein in the first direction, a length of each of the second transmission sections gradually decreases in a direction away from the at least one second transmission line.
